Aryan Luitel

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Education

Alabama Agricultural and Mechanical University **Huntsville, AL**
B.S. in Computer Science, Junior (GPA: 3.95/4.0) ***Expected Graduation: Dec 2027***

- **Coursework:** Data Structures, Theory of Algorithms, Operating Systems, Software Engineering, Machine Learning, Introduction to Databases, Distributed Systems, Object-Oriented Programming, Computer Organization, Discrete Structures, Computer Architecture

Technical Skills

- **Programming Languages:** Python, C++, Java, JavaScript, TypeScript, SQL, MATLAB, Go
- **Frameworks & Libraries:** React, React Native, NestJS, Flask, pandas, NumPy, Polars, Streamlit, LangChain
- **Cloud & Tools:** AWS, GCP (Vertex AI), Docker, PostgreSQL, Prisma, Git, GitLab CI, CUDA, Linux

Work Experience

Software Development Engineer Intern **May 2026 – Aug 2026**
Adobe *San Jose, CA*

- Developed an **AI editing agent for Adobe Express** using **Claude Opus**, **Amazon Bedrock**, and **Retrieval-Augmented Generation (RAG)**, enabling easy template editing through natural-language commands; streamlined the workflow by translating requests directly into structured document updates.
- Improved editing reliability by approximately **%15** through 50+ evaluation scenarios, failure analysis, and **end-to-end testing**; redesigned the **Entity Component System (ECS)** workflow to help the team deliver dependable editing features with stronger regression protection.

Teaching Assistant **Aug 2024 – Present**
Alabama A&M University *Huntsville, AL*

- Identified learning challenges across 50+ student-athletes and developed individualized technical support strategies, improving understanding of **Python**, **C++**, **algorithms**, and **Calculus**.
- Collected student feedback and collaborated with academic stakeholders to prioritize support needs and improve the tutoring experience.

Intern - Quantum Computing and Machine Learning **Jan 2025 - April 2025**
National Science Foundation *Remote*

- Designed and executed **quantum computing and machine learning experiments** using **Python and Qiskit** on real superconducting quantum hardware, collecting and analyzing qubit measurement data to characterize readout errors under realistic noise conditions.
- Applied **supervised machine learning concepts** to model qubit readout as a classification problem, using **confusion matrices** to quantify asymmetric errors; built a reproducible **Jupyter pipeline** to streamline data collection, validation, and experimental analysis.

Projects

Academic Research Assistant (Python, LangGraph, OpenAI API) **May 2025**

- Built a **multi-agent research assistant** using **LangGraph** to orchestrate planning, paper retrieval, and summarization steps as a stateful graph with conditional edges and a self-critique retry loop.
- Integrated the **OpenAI API** with function calling to extract structured data from academic papers, enabling consistent comparison across sources for literature review tasks.

Real-Time Bus Arrival Tracker (Python, React, Azure) **Oct 2024**

- Developed a real-time transit data platform using Python, React, Azure, and Apache Spark to ingest and process live data for bus arrival predictions.
- Built distributed Spark pipelines and CI/CD workflows to improve processing scalability, fault tolerance, data reliability, & deployment consistency.

Leadership / Extracurricular

Fellow & Campus Ambassador (Sharing the Power Foundation) **June 2026 - Present**

AWS Certified Cloud Practitioner (Amazon Web Services) **March 2026**

Google Cloud Certified: Generative AI Leader(Google) **March 2026**

NRF Foundation Ray Greenly Scholar **Jan 2026**